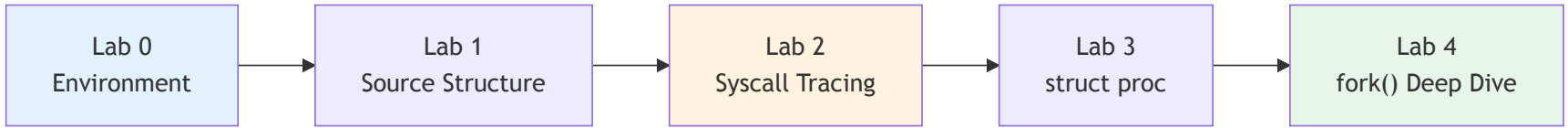
Operating Systems Lab

Week 3 – Exploring xv6 Internals

Korea University Sejong Campus, Department of Computer Science & Software

Lab Overview

- **Goal**: Navigate xv6 kernel source and trace how system calls work end-to-end
- **Duration**: ~50 minutes · 5 labs (Lab 0–4)



Lab 0: Environment Setup — Prerequisites

xv6-riscv requires a **RISC-V cross-compiler** and **QEMU emulator**.

> **QEMU** = Quick EMUlator — a software emulator that can run an entire OS for a different CPU architecture. We use it to run xv6 (built for RISC-V) on your x86/ARM laptop, without needing real RISC-V hardware.

macOS (Homebrew)

```
brew install qemu
brew install riscv64-elf-gcc
```

Ubuntu / Debian

```
sudo apt update
sudo apt install -y git build-essentia
  qemu-system-misc \
  gcc-riscv64-linux-gnu \
  binutils-riscv64-linux-gnu
```

Windows (WSL2)

```
# PowerShell (Administrator):
wsl --install -d Ubuntu
```

Restart, then in WSL Ubuntu terminal follow the Ubuntu instructions on the left.

> Work in `~/` , **not** `/mnt/c/` (very slow)

Clone xv6 (all platforms):

```
git clone https://github.com/mit-pdos/xv6-riscv.git
```

Lab 0: Verify Build & Troubleshooting

Build & Run

```
cd xv6-riscv
make qemu
```

Expected output:

```
xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$
```

Try `ls`, `echo hello` in the xv6 shell. **Exit QEMU**: press **Ctrl-A**, then **X**.

Common Issues

Problem	Fix
`riscv64 version of GCC not found`	Set `TOOLPREFIX` manually (see below)
QEMU version < 5.0	Upgrade QEMU
macOS linker errors	Use `brew install riscv64-elf-gcc` (not linux-gnu)
WSL2 "KVM not available"	Harmless warning – ignore

TOOLPREFIX – if auto-detection fails:

Lab 1: xv6 Source Structure

File	Purpose
`proc.h`	`struct proc` definition
`proc.c`	fork, exit, wait, scheduler
`syscall.c`	syscall dispatch table
`sysproc.c`	syscall handlers
`trap.c`	trap entry from user space
`usys.pl`	generates user-space stubs

Lab 2: System Call Tracing

Full path of `fork()` from user space to kernel:

```
User program: fork()
      │
      ▼
usys.S:  li a7, SYS_fork  →  ecall
      │
      ▼
trap.c:  usertrap()        ← handles all user traps
      │
      ▼
syscall.c:  syscall()      ← reads a7, dispatch table lookup
      │
      ▼
sysproc.c:  sys_fork()     ← thin wrapper
      │
      ▼
proc.c:  kfork()           ← does the actual work
```

Exercise: Add a `printf` in `sys_fork()` and rebuild to confirm you found the right spot.

Materials: `examples/skeletons/lab2_syscall_trace.patch` (TODO template) · `examples/solutions/lab2_syscall_trace.patch` (answer) **Test program**: `examples/skeletons/lab2_trace.c` (skeleton) · `examples/solutions/lab2_trace.c` (answer)

Lab 2: How to Run the Trace Exercise

Step 1 – Apply the kernel patch

```
# From the repository root:
cd xv6-riscv

# Apply the solution (or skeleton) patch
git apply ../lectures/week03/2_lab/\
examples/solutions/lab2_syscall_trace.patch
```

Step 2 – Add the test program

```
# Copy the test program into xv6 user/
cp ../lectures/week03/2_lab/\
examples/solutions/lab2_trace.c \
user/lab2_trace.c
```

Step 3 – Edit the Makefile

Open `xv6-riscv/Makefile` and find the `UPROGS` list. Add:

```
UPROGS=\
  ...
  $U/_lab2_trace\
```

Step 4 – Build and run

```
make clean && make qemu
```

At the xv6 shell prompt:

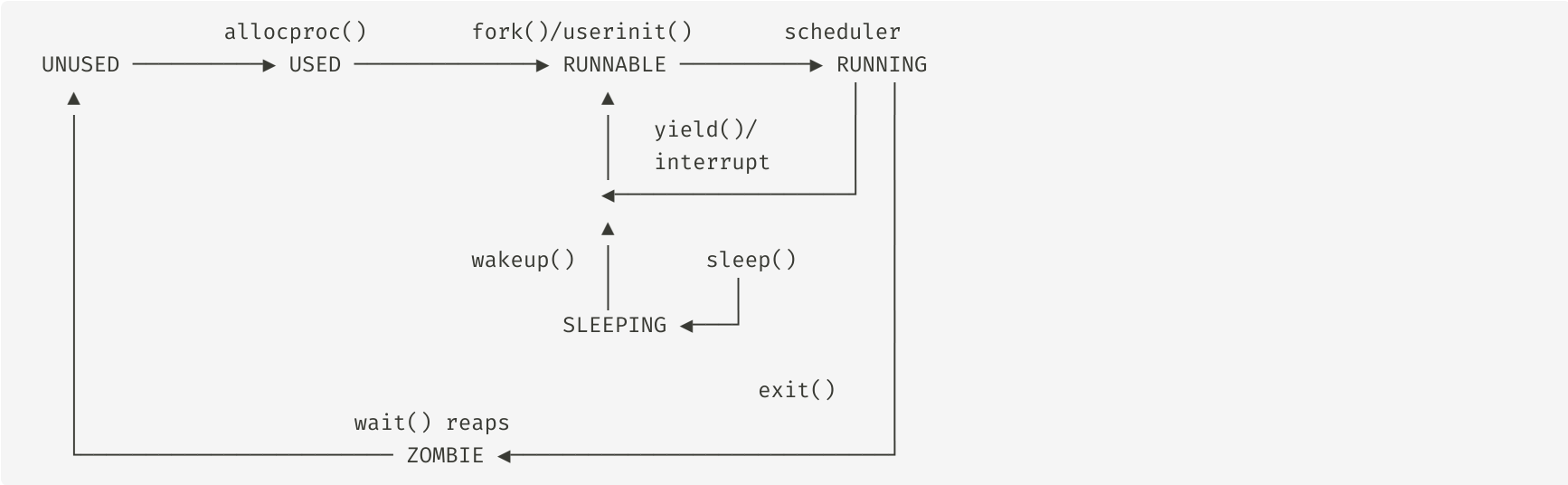
```
$ lab2_trace
```

You should see `[TRACE] sys_fork() called by ...` from the kernel.

Exit QEMU: press **Ctrl-A**, then **X**.

Lab 3: struct proc Analysis

Process state machine – defined in `kernel/proc.h`:



Key fields: `state`, `pid`, `pagetable`, `trapframe`, `context`, `ofile[]`, `parent`

- **Exercise**: Which fields change at each state transition?

Lab 3: struct proc – Full Definition

```
struct proc {
  struct spinlock lock;
  enum procstate state;        // UNUSED → USED → RUNNABLE → RUNNING → ZOMBIE
  void *chan;                  // sleep channel (if SLEEPING)
  int killed;                 // pending kill signal
  int xstate;                 // exit status for parent
  int pid;                    // process ID

  struct proc *parent;        // parent process (protected by wait_lock)

  uint64 kstack;              // kernel stack virtual address
  uint64 sz;                  // process memory size (bytes)
  pagetable_t pagetable;      // user page table
  struct trapframe *trapframe;// saved user registers (for trampoline.S)
  struct context context;     // saved kernel registers (for swtch.S)
  struct file *ofile[NOFILE]; // open file descriptors
  struct inode *cwd;          // current working directory
  char name[16];              // process name (for debugging)
};
```

Lab 4: fork() Implementation Deep Dive

```
1. allocproc()         — get new proc slot, pid, kstack, trapframe
      |
2. uvmcopy()           — copy parent's page table + memory
      |
3. Copy trapframe      — child returns from fork() too
      |
4. Set a0 = 0          — child sees fork() return 0    ★
      |
5. Copy ofile[]        — share open file descriptors
      |
6. Set parent,         — child is ready to run
   state = RUNNABLE
      |
7. Return child PID    — to parent
```

Discussion questions:

- Why does the child need its **own** trapframe copy?
- What would happen if step 4 (`a0 = 0`) were skipped?
- Why does `uvmcopy` copy **all** pages? (Hint: Week 12 – COW fork)

Key Takeaways

Concept	Key Insight
xv6	~10K lines of C – small enough to read entirely
Syscall path	user → `ecall` → `usertrap()` → `syscall()` → handler → impl
struct proc	Kernel's complete view of a process (scheduling, memory, files)
fork()	`uvmcopy` = heavy lifting; `trapframe→a0 = 0` = child return value

```
What you explored today:

Source Structure ──▶ Syscall Path ──▶ struct proc Lifecycle ──▶ fork() Implementation
```

Upcoming: threads, scheduling, and synchronization – all built on top of what you explored today.